Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/03/2006
PI Name:	Cheng, Robert K. Ph.D.		
Project Title:	Field Effects of Gravity on Lean Premixed Turbulent Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	97-HEDS-01
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	C32121A		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

This task is focused on experimental reserach to quantiry the effects of buoyancy on the mean flowfield and its impact on local and overall flame/turbulence interaction processes. The approach is to obtain detailed measurement and analysis of the velocity and scalar fields of premixed turbulent flames in microgravity and in normal, partial and enhanced gravity.

The objective of this research is to proof that buoyancy is a system parameter of laboratory and practical premixed turbulent flames. Buoyancy provides an upstream feedback mechanism that dictates the overall flame flowfield as well as local flame turbulence interaction processes. Consequently the flame structures, stabilization limits, turbulent reaction rates are directly or indirectly coupled with buoyancy.

Task Description:	The subject of our research, lean premixed turbulent flames, is an emerging environmental energy technology because these flames can be exploited to reduce the emission of oxides of nitrogen. Therefore, results of this research will benefit both combustion science and technology. Combustion engineer will have scaling information to eliminate buoyancy influences on practical systems. Experimentalists will gain new knowledge to identify and interpret buoyancy contributions to laboratory flames. Theoretical research will benefit by having a set of reference data to improve the fidelity of turbulent combustion models and to develop methods to address elliptical problems. Specifically, this research effort will generate the insight to determine significant processes through which gravity affects flame properties such as flame speed and flame structures. The results can be applied to investigate the gravity-influenced limit for premixed turbulent flames. The backward coupling effects of buoyancy (i.e., downstream effects felt upstream) is of particular importance to combustion theory because it relates to the elliptical problem that requires full time dependent treatment of upstream, wall and downstream boundary conditions. Knowledge on the back-coupling buoyancy affected processes is valuable for guiding the development of turbulent combustion models to include these effects. This study will contribute to resolving the inconsistencies that exist between experiments and theories.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Terrestrial laboratory study is an important aspect of our research that provides background information necessary to interpret data from the microgravity experiments. Due to the cancellation of the flight experiments, we re-directed our effort to design an experiment that allows us to investigate buoyancy flame coupling to support the development of models and numerical simulations. The specific goal of our FY06 activities is to collaborate with Ian Shepherd on his NASA project on “Numerical Investigation of Buoyancy and Pressure Affects in Premixed Flames.” The objective of the numerical work is to develop a methodology to address and simulate the large difference between the small length scales that characterize combustion and the large length scales that characterize the buoyant interactions. For our first problem, we considered the unstable plume of +1g and -1g ring-stabilized Bunsen type conical flames reported in our earlier experimental studies. The experiments have showed that the well-known flame tip flickering phenomenon is a direct consequence of buoyancy induced instability. As the numerical simulation of the flickering flame produces time-dependent scalar and velocity information throughout the flame flowfield, their validation requires similar types of experimental results. The regular flame flickering frequency means that the flickering cycle can provide a time base for phase-locked experimental measurements so that phased averaged velocity can be obtained for direct comparison with the numerical results. Although our ultimate goal is to investigate buoyancy effects on turbulent flames, the unstable laminar flickering flame is presently a more tractable problem to start. The flickering laminar flame experiments were performed in a 25 mm diameter burner developed for our drop-tower and parabolic flight experiments. Two-dimensional, radial, U, and axial, W, velocity data was measured by Particle Imaging Velocimetry (PIV) and phase-locked flowfield information was obtained by triggered the PIV using a schlieren-based flame motion sensor. For each flame, mean velocity vectors were obtained at four or eight phase angles during the cycle. Examination of the phase-locked and time averaged velocity results from four flames clearly demonstrates the vortex ‘pumping’ mechanism that generates the flame flicker. The effect of this process is also discernible from a comparison of the experimental and numerical phase-averaged axial centerline velocity profiles. This study has shown that the phase-locked PIV technique provides valuable insights to support numerical simulation of the buoyancy induced large scale unstable flow phenomenon. More importantly, the vortex-pairing process shown by the simulation helps to explain some of the ambiguities found in the phased-averaged PIV data. This shows that a synergistic experimental and simulation approach is very beneficial to furthering the understanding of field effects of gravity to combustion. We plan to continue with further analyses of the experimental and numerical data obtained for the laminar pulsating flames. In addition, phase-locked PIV measurements will be made to investigate vortex-pairing and also in turbulent flames to investigate the effect of turbulence in plume vortex formation.
Bibliography Type:	Description: (Last Updated: 07/09/2019)
Papers from Meeting Proceedings	Shepherd IG, Cheng RK, Day MS. "The Dynamics of Flame Flicker in Conical Premixed Flames." 31st International Combustion Symposium, Heidelberg, Germany, August 6-11, 2006. Poster Paper. 31st International Combustion Symposium, Heidelberg, Germany, August 6-11, 2006. , Aug-2006